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Predictive AI for Active Aerodynamics Efficiency and Driver Anomaly Analysis

A Formal Technical Report — Two-Phase Research Project

*Phase 1: Anomaly Detection & Aero State Classification / Phase 2:
Anticipatory Aero Mode Prediction*

Submitted by: Anas Norani · Hanan Majeed · Maha Mohsin

Program: BS Data Science — SEECS, NUST

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Phase 1 | RF AUC-ROC = 1.000

Phase 2 | LR AUC-ROC = 0.963 (LOCO)

4 F1 Circuits | 8 Models | 5 Horizons

159,712 Telemetry Rows

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Abstract

This report presents a comprehensive two-phase machine learning study applied to Formula 1 car telemetry data. **Phase 1** investigates driver anomaly detection and current aerodynamic (aero) state classification using a single-race dataset from the 2026 Japanese Grand Prix at Suzuka (63,673 rows). Using Isolation Forest, K-Means clustering, Logistic Regression, and Random Forest, we establish that the current aero mode is perfectly classifiable from instantaneous sensor readings (Random Forest AUC-ROC = 1.000), while anomalous driving windows — predominantly maximum-braking and gear-shift events — exhibit statistically significant feature deviations (Cohen's d up to -1.41). **Phase 2** extends this to the harder problem of *anticipatory prediction*: given a 5-second history of car telemetry, predict the aero mode **1 second into the future** on circuits the model has never trained on. We benchmark eight models under Leave-One-Circuit-Out (LOCO) evaluation across four circuits (Monaco, Monza, Silverstone, Suzuka; 159,712 rows). Logistic Regression on instantaneous features achieves the best mean AUC-ROC of **0.963**, while the Transformer achieves the strongest deep-learning performance (**0.911**). A critical finding is the catastrophic collapse of the Gated Recurrent Unit (GRU) at the Monaco circuit (AUC-ROC = 0.728, F1-Xmode = 0.005), caused by circuit-level speed-distribution memorisation — a failure mode also observable in the Temporal Convolutional Network at Suzuka (AUC-ROC = 0.537). All models fail beyond a 2.5-second horizon, constraining practical systems to sub-second anticipation windows. The unifying insight across both phases is that models which encode circuit-specific speed baselines fail to generalise — a property that directly connects Phase 1's anomaly detection to Phase 2's cross-circuit breakdown.

Keywords: Formula 1 telemetry, active aerodynamics, anomaly detection, anticipatory prediction, leave-one-circuit-out, time-series classification, Transformer, GRU, circuit memorisation.

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1. Introduction and Motivation

The 2026 Formula 1 technical regulations (FIA Art. 3.10) introduce mandatory Active Aerodynamic systems that physically reconfigure a car's wing geometry between two operating modes:

- **Z-Mode (High Downforce):** Wings fully deployed to maximise mechanical grip through corners.
- **X-Mode (Low Drag):** Wings retracted to reduce aerodynamic resistance on fast straights.

The transition rule is governed by a fixed threshold: the car enters X-Mode when its speed exceeds **240 km/h** while in **gear 6 or higher**. While the optimal mode at any given moment is determined by physics, the challenge is one of **timing**: actuation, sensing, and onboard decision pipelines all incur latency on the order of 0.1–0.3 seconds. A control system that reacts only when the threshold is already crossed operates too late to be fully effective.

This project investigates whether machine learning can solve this timing problem in two steps:

Phase 1 — Before predicting the future, we must understand the present. Using data from the 2026 Suzuka Grand Prix, we ask: (i) Can the current aero mode be identified from sensor data? (ii) Are there detectable anomalies in driver behaviour? (iii) What do different driving regimes look like statistically?

Phase 2 — Building on Phase 1's feature understanding, we ask the harder question: can a model predict the aero mode **1 second before** the car reaches that state, on circuits it has never seen during training?

2. Dataset Description

All telemetry data was collected via the **FastF1** open-source Python library from official FIA timing feeds at approximately 10 Hz (10 samples per second per car).

2.1 Phase 1 Dataset

Property	Value
Race	2026 Japanese Grand Prix
Circuit	Suzuka International Racing Course
Drivers	Max Verstappen (RB), Isack Hadjar (RB)
Total rows	63,673
Sampling rate	~10 Hz
X-Mode proportion	34.7%
Z-Mode proportion	65.3%

2.2 Phase 2 Multi-Circuit Dataset

Circuit	Race	Rows	X-Mode %	Speed Profile
Monaco	2025 Monaco GP	39,881	13.3%	Slow street (max ~220 km/h)
Monza	2025 Italian GP	31,098	56.9%	High-speed power (max ~365 km/h)
Silverstone	2025 British GP	25,060	38.9%	Mixed (max ~310 km/h)
Suzuka	2026 Japanese GP	63,673	34.7%	Technical (max ~330 km/h)
Total		159,712	~36%	

The four circuits were chosen to maximise diversity in speed profiles, corner types, and X-Mode prevalence — intentionally creating a challenging generalisation benchmark. Monaco's X-Mode rate of only 13.3% (versus Monza's 56.9%) is the single largest source of domain shift in the study.

2.3 Aero Mode Label Definition

```
y[t] = 1 (X-Mode) if Speed[t] >= 240 km/h AND nGear[t] >= 6
y[t] = 0 (Z-Mode) otherwise
```

In Phase 2, the label uses the **future** time step: `y[t+H]`, where H is the prediction horizon. The model never sees future data in its input window.

3. Phase 1 — Driver Anomaly Detection and Aero State Classification

3.1 Feature Engineering

Starting from raw FastF1 channels (Speed, RPM, nGear, Throttle, Brake, X/Y/Z position), the following additional features were computed:

Feature	Formula / Source	Rationale
Acceleration (m/s ²)	$\Delta \text{Speed} / \Delta \text{time}$	Longitudinal dynamics
Elevation_Delta (m)	$\Delta Z / \Delta \text{time}$	Track gradient signal
Kinetic_Energy_MJ	$\frac{1}{2} \times 798 \text{ kg} \times (\text{Speed}/3.6)^2 / 10^6$	Physics-grounded energy
Longitudinal_Force_N	$798 \times \text{Acceleration} / 3.6$	Force proxy
High_Speed_Zone	1 if $\text{Speed} \geq 240$ AND $\text{nGear} \geq 6$	Aero mode indicator
Heavy_Braking	1 if $\text{Brake} \geq 50\%$ AND $\text{Accel} \leq -5 \text{ m/s}^2$	Braking event flag
Gear_Shift_Active	1 if consecutive gear change within 0.5 s	Gear transition flag

Descriptive statistics (Phase 1 dataset, n=63,673):

Feature	Mean	Std	Min	25th %	Median	75th %	Max	Skewness
Speed (km/h)	218.2	65.4	58.0	173.5	216.0	272.0	347.0	-0.21
RPM	10,619	811	6,252	10,175	10,741	11,175	12,708	-0.96
nGear	5.48	1.90	1	4	5	7	8	-0.40
Throttle (%)	62.5	42.5	0	13.0	90.6	100.0	104.0	-0.47
Acceleration (m/s ²)	0.01	4.85	-28.0	-0.81	0.00	1.40	28.0	-0.81
Elevation_Delta (m)	0.00	4.34	-82.0	-1.00	0.01	1.13	108.5	-0.07
Kinetic_Energy (MJ)	1.60	0.86	0.10	0.93	1.44	2.28	3.71	+0.31

3.2 Anomaly Detection — Isolation Forest

Method: Isolation Forest with contamination=5%, 100 estimators, random_state=42. Features used: Speed, RPM, nGear, Throttle, Brake, Acceleration, Elevation_Delta.

Result: Approximately **5% of rows (≈3,184 samples)** flagged as anomalous.

Feature comparison — anomalous vs. normal windows:

Feature	Anomalous Mean	Normal Mean	Cohen's d	Effect Size	MW p-value
Acceleration (m/s ²)	-6.20	+0.34	-1.41	Large	<0.001
Throttle (%)	22.7	64.6	-1.01	Large	<0.001
Brake	0.686	0.157	+1.44	Large	<0.001
Gear_Shift_Active	0.711	0.022	+3.96	Large	<0.001
Heavy_Braking	0.290	0.000	+2.86	Large	<0.001

All differences are highly statistically significant (Mann-Whitney U, $p < 0.001$ for all). The anomaly signature is: **near-zero throttle, hard braking, strong negative acceleration, and active gear shifting** — characteristic of late-braking traction limit events at technical corners.

Association with aero mode: Chi-squared test ($\chi^2 = 817.4$, $p = 9.0 \times 10^{-180}$, OR = 0.18 [95% CI: 0.16–0.20]) confirms anomalous windows are strongly associated with Z-Mode. Anomalous windows are only **18% as likely** to be in X-Mode as normal windows — anomalies occur almost exclusively during braking events, not on straights.

Geographical analysis: Anomalies mapped to GPS coordinates cluster at Suzuka's known technical sections: the Esses complex (Turns 3–7), Degner curves (Turns 8–9), the Hairpin (Turn 11), Spoon Curve (Turn 13), and the 130R–Casio chicane approach.

3.3 Driving Zone Clustering — K-Means

Method: K-Means with $k=4$, determined by elbow method on within-cluster sum of squares. Features normalised before clustering: Speed, nGear, Acceleration, Brake, Throttle.

Statistical validation: Kruskal-Wallis H-test on Speed across four clusters: **$H = 34,439$, $p < 0.001$, $\eta^2 = 0.555$** .

Cluster membership explains **55.5% of variance in Speed** — confirming clusters are physically meaningful. All six pairwise Dunn comparisons are significant at $p < 0.001$ (Bonferroni corrected).

Cluster characterisation:

Cluster	Dominant Behaviour	Mean Speed (km/h)	Mean Throttle (%)	Aero Mode
0	High-speed aero phase	295 ± 28	98 ± 4	X-Mode (dominant)
1	Medium cornering	210 ± 35	71 ± 22	Z-Mode
2	Heavy braking and entry	165 ± 42	12 ± 19	Z-Mode
3	Slow hairpin approach	112 ± 38	8 ± 15	Z-Mode

3.4 Aero State Classification

Method: Both Logistic Regression (L2, C=1.0) and Random Forest (100 trees, balanced class weights) trained on instantaneous features. 80/20 stratified train/test split. Evaluated using bootstrap confidence intervals (10,000 resamples, seed=42).

Results:

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	AUC 95% CI
Random Forest	1.000	1.000	1.000	1.000	1.000	[1.000, 1.000]
Logistic Regression	0.979	0.881	1.000	0.937	0.9997	[0.9993, 0.9999]

Interpretation: Both models achieve near-perfect performance because the aero mode label is defined by thresholds on Speed and nGear — features that are present in the input. This result validates the feature set and confirms the task is physically sound, but it does not demonstrate genuine predictive capability. This directly motivates Phase 2, where the label is shifted H=10 steps into the future and the model cannot simply look up the answer.

DeLong AUC comparison: $\Delta\text{AUC} = 0.0003$, $z = -2.06$, $p = 0.039$. RF is statistically but not practically superior to LR at the boundary. McNemar test: RF wins on 274 edge-case samples vs. LR's 0 ($\chi^2 = 272$, $p \approx 0$), indicating RF handles the non-linear speed/gear boundary more precisely.

3.5 Statistical Analysis

3.5.1 Normality Tests

All 11 features were tested with Shapiro-Wilk (subsample $n=5,000$) and D'Agostino-Pearson K^2 . **All features are significantly non-normal ($p < 0.001$ for both tests on all features).** This rules out parametric methods and motivates bootstrap CIs and non-parametric tests throughout.

Feature	SW Statistic	SW p	D-P Statistic	D-P p	Verdict
Speed	0.978	<0.001	4,541.6	<0.001	Non-normal
RPM	0.961	<0.001	9,037.1	<0.001	Non-normal
nGear	0.928	<0.001	3,247.7	<0.001	Non-normal
Throttle	0.768	<0.001	358,737	<0.001	Non-normal
Acceleration	0.796	<0.001	16,482.7	<0.001	Non-normal
Elevation_Delta	0.576	<0.001	24,598.6	<0.001	Non-normal
Kinetic_Energy	0.962	<0.001	11,485.9	<0.001	Non-normal

3.5.2 Driver Comparison — Verstappen vs. Hadjar

Test	Statistic	p-value	Effect Size	Interpretation
Mann-Whitney U on Speed	U = 0.0	<0.001	Cohen's d = 0.042	Significant but small
Chi-squared on aero deviation rate	$\chi^2 = 73.1$	<0.001	Cramér's V = 0.113	Small association

Both differences are statistically significant but practically small ($d < 0.1$, $V < 0.2$). Verstappen shows marginally higher average Speed and a slightly different X-Mode entry rate. Driver differences exist but are not large enough to warrant driver-specific models for Phase 2.

3.5.3 Permutation Feature Importance

Rank	Feature	Importance	Std	95% CI	Significant
1	nGear	0.167	0.006	[0.155, 0.180]	YES
2	Elevation_Delta	0.162	0.004	[0.156, 0.170]	YES
3	Brake	0.081	0.003	[0.075, 0.086]	YES
4	Speed	0.0004	0.0001	[0.000, 0.0005]	YES
5–18	All other features	≈ 0.000	≈ 0	—	NO

Key observation: nGear outranks Speed as the most important feature. This reflects that nGear is the harder constraint: maintaining high gear requires both high speed *and* appropriate track position, whereas speed alone can be high in many configurations. Elevation_Delta's high importance is Suzuka-specific — the circuit's varied terrain creates reliable gear-zone boundaries. This finding directly informs the feature interpretation in Phase 2 (Section 4.7).

4. Phase 2 — Anticipatory Aero Mode Prediction

4.1 Problem Formulation

The anticipatory prediction task is formally defined as:

Input: A sliding window of $W=50$ time steps ending at time t :

$X = [x_{\{t-49\}}, x_{\{t-48\}}, \dots, x_t]$ with shape $(50, 12)$

Target: Aero mode label $H=10$ steps into the future:

$y[t+10] = 1$ if $\text{Speed}[t+10] \geq 240$ AND $\text{nGear}[t+10] \geq 6$

The **label index is strictly greater than the last input index**. The future state that defines the label is never available to the model. This formulation resolves the degeneracy identified in Phase 1 — the model cannot simply recover the threshold rule from its inputs.

Window parameters:

Parameter	Value	Meaning
W	50 samples	5-second input window
Primary H	10 samples	1-second look-ahead
Horizon sweep	$H \in$	0.1s to 5.0s ahead
F	12 features	See Section 4.2

Feature set (F=12):

Category	Features
Raw telemetry	Speed, RPM, nGear, Throttle, Brake, X, Y, Z
Physics-derived	Acceleration, Elevation_Delta, Kinetic_Energy_MJ, Longitudinal_Force_N

Note: Engine_Load (= $\text{RPM} \times \text{Throttle} / 100$) is a fabricated proxy absent from raw telemetry and is intentionally excluded.

Evaluation protocol — Leave-One-Circuit-Out (LOCO):

Fold	Train Circuits	Test Circuit
1	Monza + Silverstone + Suzuka	Monaco
2	Monaco + Silverstone + Suzuka	Monza
3	Monaco + Monza + Suzuka	Silverstone
4	Monaco + Monza + Silverstone	Suzuka

StandardScaler is fit on **training windows only** — no test-circuit statistics leak into normalisation.

4.2 Models and Experimental Setup

Model ladder (ordered by temporal capacity):

Model	Architecture	Parameters	Temporal Reasoning
LR-instant	Logistic Regression on x_t only	~12	None
RF-instant	Random Forest on x_t only	~50k	None
RF-lag	Random Forest on flattened 50×12 window	~50k	Weak (explicit lags)
CNN	Causal Conv1D × 3 blocks, global avg pooling	~136k	Local patterns
CausalLSTM	2-layer unidirectional LSTM, hidden=128	~213k	Long-range recurrent
CausalGRU	2-layer unidirectional GRU, hidden=128	~162k	Long-range recurrent
TCN	4 dilated causal residual blocks (d=1,2,4,8)	~152k	Fixed receptive field
CausalTransformer	2-layer encoder, 4 heads, causal mask	~68k	Self-attention

Training configuration (all deep models):

Hyperparameter	Value
Loss function	Focal Loss ($\gamma=2.0$, $\alpha=1 - \text{pos_rate}$ per fold)
Optimiser	AdamW ($\text{lr}=3 \times 10^{-4}$, $\text{weight_decay}=10^{-4}$)
LR schedule	Cosine annealing + ReduceOnPlateau (patience=5)
Batch size	512
Dropout	0.2 (0.1 for Transformer)
Early stopping	Patience=15, monitors AUC-PR on 10% val split
Max epochs	100
Gradient clipping	$\text{norm} \leq 1.0$
Reproducibility	seed=42
Hardware	NVIDIA P100 GPU, 16 GB VRAM (Kaggle cloud)

4.3 Primary Results at H=10 (1 Second Ahead)

Table 1 — Mean ± std across 4 LOCO folds at H=10 (primary horizon)

Model	Temporal	F1-Macro	AUC-ROC	AUC-PR	σ AUC-ROC
LR-instant	No	0.906 $\pm$ 0.022	0.963 $\pm$ 0.021	0.881 $\pm$ 0.067	0.021
RF-instant	No	0.872 $\pm$ 0.050	0.959 $\pm$ 0.011	0.864 $\pm$ 0.099	0.011
RF-lag	Weak	0.794 $\pm$ 0.148	0.945 $\pm$ 0.026	0.815 $\pm$ 0.162	0.026
Transformer	Yes	0.779 $\pm$ 0.115	0.911 $\pm$ 0.055	0.783 $\pm$ 0.096	0.055
LSTM	Yes	0.756 $\pm$ 0.139	0.878 $\pm$ 0.085	0.729 $\pm$ 0.121	0.085
GRU	Yes	0.768 $\pm$ 0.179	0.877 $\pm$ 0.104	0.724 $\pm$ 0.273	0.104
TCN	Yes	0.785 $\pm$ 0.117	0.835 $\pm$ 0.201	0.755 $\pm$ 0.167	0.201
CNN	Yes	0.748 $\pm$ 0.093	0.818 $\pm$ 0.111	0.712 $\pm$ 0.148	0.111

Key observations from Table 1:

- No deep model surpasses LR-instant.** The gap between LR-instant and the best deep model (Transformer) is 0.052 AUC-ROC points. This is not a marginal difference — it is statistically confirmed (Section 4.9).
- Standard deviation is diagnostic.** LR-instant's $\sigma = 0.021$ versus GRU's $\sigma = 0.104$ and TCN's $\sigma = 0.201$ reflects catastrophic per-fold failure, not uniform underperformance. A model with high σ is not consistently bad — it is excellent on some circuits and collapses on others.
- Transformer is the most circuit-robust deep model** ($\sigma = 0.055$), outperforming all other deep architectures on both mean AUC-ROC and cross-circuit stability.

4.4 Per-Circuit LOCO Breakdown

Table 2 — AUC-ROC per held-out circuit at H=10

Model	Monaco	Monza	Silverstone	Suzuka	Mean $\pm$ std
LR-instant	0.980	0.932	0.970	0.969	0.963 $\pm$ 0.021
RF-instant	0.955	0.963	0.972	0.945	0.959 $\pm$ 0.011
RF-lag	0.907	0.947	0.963	0.961	0.945 $\pm$ 0.026
Transformer	0.956	0.871	0.961	0.857	0.911 $\pm$ 0.055
LSTM	0.927	0.863	0.957	0.765	0.878 $\pm$ 0.085
GRU	0.728	0.896	0.968	0.917	0.877 $\pm$ 0.104
TCN	0.944	0.890	0.968	0.537	0.835 $\pm$ 0.201
CNN	0.811	0.821	0.956	0.684	0.818 $\pm$ 0.111

Table 3 — F1-Macro per held-out circuit at H=10

Model	Monaco	Monza	Silverstone	Suzuka	Mean $\pm$ std
LR-instant	0.911	0.873	0.935	0.905	0.906 $\pm$ 0.026
RF-instant	0.803	0.902	0.934	0.851	0.872 $\pm$ 0.059
RF-lag	0.544	0.838	0.923	0.870	0.794 $\pm$ 0.168
Transformer	0.806	0.802	0.914	0.596	0.779 $\pm$ 0.133
LSTM	0.770	0.832	0.895	0.528	0.756 $\pm$ 0.157
GRU	0.465	0.874	0.919	0.812	0.768 $\pm$ 0.196
TCN	0.760	0.849	0.924	0.609	0.785 $\pm$ 0.139
CNN	0.784	0.772	0.841	0.594	0.748 $\pm$ 0.106

Table 4 — F1-Xmode (X-Mode class only) per held-out circuit at H=10

Model	Monaco	Monza	Silverstone	Suzuka
LR-instant	0.849	0.903	0.922	0.879
RF-instant	0.650	0.920	0.917	0.797
RF-lag	0.157	0.863	0.904	0.824
Transformer	0.660	0.848	0.896	0.376
LSTM	0.594	0.861	0.877	0.256
GRU	0.005	0.901	0.903	0.740
TCN	0.572	0.881	0.908	0.409
CNN	0.619	0.838	0.826	0.376

Critical failure analysis:

The **GRU at Monaco** is the most severe failure in the study. With AUC-ROC = 0.728 and F1-Xmode = **0.005**, the model predicts X-Mode on fewer than 0.5% of samples where it is required. This is not a weak classifier — it is essentially a constant Z-Mode predictor. On the same circuit with the same data, Transformer achieves AUC-ROC = 0.956 and LR-instant achieves 0.980. The failure is model-specific, not data-specific.

The **TCN at Suzuka** (AUC-ROC = 0.537, near random chance) is a second, independent failure mode. Unlike GRU's circuit-speed collapse, TCN's failure at Suzuka appears linked to the circuit's irregular gear/throttle cadence at high speed, which is incompatible with TCN's fixed dilated receptive field trained predominantly on smoother European straight-corner patterns.

4.5 Horizon Sweep Analysis

Table 5 — Mean AUC-ROC vs. prediction horizon (4 LOCO folds)

Model	H=1 (0.1s)	H=5 (0.5s)	H=10 (1s)	H=25 (2.5s)	H=50 (5s)
LR-instant	0.999	0.993	0.963	0.786	0.471
RF-instant	0.999	0.993	0.959	0.777	0.562
RF-lag	0.999	0.985	0.945	0.745	0.466
Transformer	0.999	0.987	0.911	0.691	0.595
GRU	0.999	0.986	0.877	0.682	0.505

Table 6 — GRU per-circuit AUC-ROC across all horizons

Circuit	H=1	H=5	H=10	H=25	H=50
Monaco	0.9996	0.9894	0.728	0.693	0.804
Monza	0.9993	0.9777	0.896	0.635	0.296
Silverstone	0.9994	0.9907	0.968	0.818	0.565
Suzuka	0.9992	0.9873	0.917	0.582	0.356

Table 7 — Transformer per-circuit AUC-ROC across all horizons

Circuit	H=1	H=5	H=10	H=25	H=50
Monaco	0.9995	0.9951	0.956	0.791	0.825
Monza	0.9995	0.9860	0.871	0.696	0.397
Silverstone	0.9993	0.9912	0.961	0.849	0.687
Suzuka	0.9991	0.9758	0.857	0.628	0.470

Horizon regime analysis:

Three regimes are clearly visible across Tables 5–7:

Trivial regime ($H \leq 5$, $\leq 0.5s$): All models exceed AUC-ROC 0.975. High temporal autocorrelation makes the task nearly deterministic — knowing the car's current state is sufficient to predict with high confidence what state it will be in 0.5 seconds later. These results are not informative of model capability.

Meaningful anticipation regime ($H = 10$, $1s$): Performance diverges substantially. The gap between best (LR-instant, 0.963) and worst (CNN, 0.818) exceeds 14 AUC-ROC points. Circuit-specific failures emerge (GRU at Monaco drops from 0.989 at $H=5$ to 0.728 at $H=10$). This is the primary evaluation

horizon.

Long-horizon breakdown ($H \geq 25$, $\geq 2.5s$): All models degrade to below AUC-ROC 0.79. At $H=50$ (5 seconds ahead), LR-instant achieves only 0.471 (below chance) and GRU Monza achieves only 0.296. Only the Transformer retains marginally above-chance performance (0.595 at $H=50$), suggesting its self-attention mechanism captures some minimal long-range dependency.

Engineering implication: A practical active-aero control system must operate within approximately 1-second anticipation. No current machine-learning method is reliable enough for predictive control beyond 2.5 seconds.

4.6 Transition Window Analysis

Mode changes are the most safety-critical and operationally relevant prediction events. We isolate **transition windows**: the ± 5 samples (± 0.5 seconds) centred on each true mode-change event.

Table 8 — F1-Macro: transition vs. stable windows at $H=10$

Circuit	Model	N Transition	F1 Transition	F1 Stable	Δ F1
Monaco	Transformer	4,304	0.496	0.902	−0.406
Monaco	GRU	4,304	0.328	0.480	−0.152
Monza	Transformer	5,958	0.551	0.865	−0.314
Monza	GRU	5,958	0.615	0.943	−0.327
Silverstone	Transformer	4,150	0.659	0.970	−0.311
Silverstone	GRU	4,150	0.674	0.973	−0.299
Suzuka	Transformer	8,809	0.554	0.599	−0.045
Suzuka	GRU	8,809	0.575	0.860	−0.285
MEAN	Transformer	—	0.565	0.834	−0.269
MEAN	GRU	—	0.548	0.814	−0.266

Interpretation: A mean F1 drop of ≈ -0.27 at transition windows is observed for both models across all circuits. This is a **fundamental property of the anticipatory task**: at the ± 0.5 second boundary of a mode change, the car's telemetry has not yet fully committed to the new regime, making it impossible for any model to predict with high confidence. This result holds even on high-performing circuits (GRU at Silverstone: stable F1 = 0.973, transition F1 = 0.674).

Monaco anomaly: GRU's stable-window F1 at Monaco is already only 0.480 (the circuit collapse described in Section 4.4 depresses the baseline). Its transition F1 of 0.328 represents a total breakdown. Transformer, by contrast, achieves 0.902 on stable Monaco windows despite the same distribution shift.

4.7 Feature Interpretability — Integrated Gradients

Method: Integrated Gradients (IG) with baseline = zero tensor, 50 interpolation steps, computed on 300 randomly sampled test windows per fold.

Table 9 — Feature importance: GRU vs. Transformer (Monaco fold, H=10)

Feature	GRU IG rank	GRU IG value	Transformer IG rank	Transformer IG value	--- --- --- ---
Longitudinal_Force_N	#1	0.099	#5	0.037	Acceleration
Throttle	#3	0.085	#1	0.120	#2
RPM	#9	0.038	#4	0.078	#6
Elevation_Delta	#7	0.052	#5	0.074	#8
X (position)	#10	0.033	#7	0.026	#3
Y (position)	#11	0.028	#8	0.040	#9
Z (position)	#12	0.021	#9	0.019	#4
			#10	0.033	#5
			#11	0.015	#6
			#12	0.013	#7

Mechanistic interpretation:

GRU's top-ranked features at Monaco are **dynamics features** — Longitudinal Force and Acceleration describe *how fast* the car is changing speed. During training on Monza and Silverstone, these signals fire strongly in the seconds before a fast straight (high positive acceleration approaching X-Mode). At Monaco, this pattern never reaches the trained threshold — the circuit's lower speed envelope means "strong positive acceleration to 250 km/h" never occurs. GRU's prediction engine is calibrated to a speed envelope it never encounters at Monaco.

Transformer's top-ranked features are **instantaneous state features** — Throttle, Speed, nGear, RPM describe *what the car is doing right now*. These signals transfer across circuits because the X-Mode threshold rule (Speed ≥ 240 AND nGear ≥ 6) is identical at Monaco, Monza, Silverstone, and Suzuka. A car approaching that threshold looks the same instantaneously regardless of the circuit's overall speed level.

This contrast mechanistically explains the performance gap in Table 2.

Time-step attribution (Monaco fold):

Model	Late-window mean IG (steps 40–49)	Early-window mean IG (steps 0–9)	Ratio	--- ---
GRU	0.081	0.058	1.40×	
Transformer	0.050	0.025	2.00×	

Both models weight recent time steps more heavily than early ones. Transformer is more strongly focused on recent context (2× ratio), consistent with last-token classification attending primarily to its immediate neighbourhood. GRU's smaller ratio suggests its hidden state retains some early-window context — the compressed speed trajectory from training circuits that misfires at Monaco.

4.8 Transformer Attention Maps

Average self-attention weights (50×50 matrix) are extracted per fold by running each encoder layer with `need_weights=True` and averaging over batch, layers, and test samples.

Key finding: The attention topology is **consistent across all four LOCO folds**. Monaco and Silverstone produce qualitatively similar average attention matrices: strongest weights along the diagonal (current time step attending to itself) and the immediately preceding 5–10 steps, with a secondary cluster at the very first time step (long-range context). There is **no circuit-specific reorganisation** of attention patterns — the model does not learn to attend differently at Monaco versus Monza. This is the attention-map confirmation that Transformer's internal representation is less circuit-dependent than GRU's hidden state.

4.9 Statistical Significance

With only $n=4$ LOCO folds, standard paired tests (Wilcoxon signed-rank, DeLong) cannot achieve $p < 0.125$ (two-tailed minimum with $n=4$). We therefore report **bootstrap 95% confidence intervals** on mean pairwise AUC-ROC differences (LR-instant minus comparator; 10,000 resamples of fold-level observations, seed=42).

Table 10 — Bootstrap CI for AUC-ROC gap vs. LR-instant at H=10

Comparator	Per-fold deltas (Mo / Mz / Si / Su)	Mean Gap	95% CI	Interpretation
RF-instant	[+0.025, -0.031, -0.002, +0.023]	+0.004	[-0.017, +0.024]	CI includes zero — not significant
RF-lag	[+0.073, -0.015, +0.007, +0.007]	+0.018	[-0.010, +0.057]	CI includes zero — not significant
Transformer	[+0.024, +0.061, +0.009, +0.111]	+0.051	[+0.017 , +0.090]	CI excludes zero — significant
LSTM	[+0.053, +0.069, +0.013, +0.204]	+0.085	[+0.027 , +0.166]	CI excludes zero — significant
GRU	[+0.252, +0.036, +0.003, +0.052]	+0.086	[+0.015 , +0.198]	CI excludes zero — significant
TCN	[+0.036, +0.041, +0.002, +0.431]	+0.128	[+0.012 , +0.332]	CI excludes zero — significant
CNN	[+0.169, +0.111, +0.015, +0.285]	+0.145	[+0.053 , +0.241]	CI excludes zero — significant

Summary of statistical findings:

- LR-instant is **not statistically distinguishable** from RF-instant and RF-lag by AUC-ROC alone (CIs include zero). The three classical models are statistically equivalent in aggregate performance.
- LR-instant is **statistically significantly better** than all five deep learning models (all AUC-ROC CIs exclude zero). The advantage is not a sampling artefact of the four circuits chosen.

- The wide CIs for GRU ([+0.015, +0.198]) and TCN ([+0.012, +0.332]) reflect **circuit-specific catastrophic failure** rather than uniform underperformance — a model that scores +0.252 at Monaco and +0.003 at Silverstone is not consistently bad.
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5. Discussion

5.1 Why the Simplest Model Wins

The result that Logistic Regression on 12 instantaneous features (≈ 12 parameters) outperforms a GRU with 162,000 parameters is counterintuitive. The explanation is **circuit memorisation**.

Under LOCO evaluation, a model that encodes circuit-specific patterns from training data will fail when those patterns do not transfer to the test circuit. Consider what each model memorises:

RF-lag receives the full 5-second speed history. It learns: *"When speed has been rising through 220–250 km/h over the past 5 seconds, X-Mode is imminent."* This rule is correct at Monza (where speed regularly reaches 340 km/h) and Silverstone (280 km/h). At Monaco, where maximum speed barely exceeds 220 km/h, this rule never fires — and RF-lag collapses to $F1\text{-Xmode} = 0.157$.

GRU compresses this same speed trajectory into a hidden state. After training on fast circuits, the hidden state has learned: *"If the speed trajectory has been climbing steeply, activate X-Mode prediction."* At Monaco, no sample looks like any training sample in terms of speed progression. GRU predicts Z-Mode for 99.5% of Monaco's test samples, yielding $F1\text{-Xmode} = 0.005$.

LR-instant has no memory. It evaluates: *"Is the current Speed approaching 240 km/h AND current nGear approaching 6?"* The approach to this threshold looks similar regardless of which circuit the car is on. Hence LR remains robust across all four circuits ($\sigma = 0.021$ vs GRU's $\sigma = 0.104$).

Transformer survives better than GRU because its attention mechanism focuses on instantaneous state features (Throttle, Speed, nGear — confirmed by Table 9) rather than accumulated dynamics. Attention is less sensitive to absolute speed magnitude than GRU's compressed trajectory representation.

5.2 Connection Between Phase 1 and Phase 2

Phase 1 established that the X-Mode threshold rule is recoverable from instantaneous features with perfect accuracy. Phase 2 shows that shifting this task to 1-second look-ahead on unseen circuits is hard precisely because models that encode circuit-specific patterns from training data fail.

This connection is direct: the anomalies detected by Isolation Forest in Phase 1 are braking events that deviate from the circuit's "normal" driving pattern ($OR = 0.18$ vs. X-Mode). GRU's hidden state represents the same circuit-specific "normal" speed profile. Monaco's lower speed envelope is to GRU what an anomalous braking event is to the Isolation Forest at a fast circuit — a statistical outlier from the learned distribution.

The models that survive cross-circuit evaluation (LR-instant, Transformer) are the models whose representations are not grounded in circuit-specific baselines. This is the unifying principle across both phases.

5.3 Practical Engineering Implications

- 1. Control system design:** Real-time active aero systems should use $H=10$ (1-second) anticipation. The trivial regime ($H \leq 5$) provides no meaningful advance notice, and the breakdown regime ($H \geq 25$) is unreliable.
- 2. Model selection:** For deployment on unseen circuits, LR-instant is the safest choice (lowest σ , highest mean AUC-ROC). If a deep model is required for other reasons, Transformer is the only deep architecture with demonstrated cross-circuit robustness.
- 3. Feature design:** The most important features are instantaneous state (Throttle, Speed, nGear, RPM). Adding circuit-adaptive normalisation — expressing Speed as "fraction of circuit maximum" rather than absolute km/h — would likely close the gap between LR-instant and deep models.

6. Conclusions

This project demonstrates two connected findings:

Phase 1: The current aerodynamic mode is perfectly classifiable from instantaneous telemetry features (RF AUC-ROC = 1.000). This validates the feature set but establishes a ceiling, not a contribution. Anomalous driving windows ($\approx 5\%$ of data) exhibit large-effect deviations from normal patterns (Cohen's d up to -1.41), predominantly corresponding to braking events at technical corners.

Phase 2: Anticipatory prediction of aero mode 1 second ahead on unseen circuits is feasible but significantly harder. Logistic Regression (AUC-ROC = 0.963) is the most reliable model; no deep learning architecture surpasses it. The critical failure mode — circuit memorisation — affects every model that encodes absolute speed levels from training circuits. GRU collapses at Monaco (AUC-ROC = 0.728), while the Transformer maintains robustness (0.956) by focusing on instantaneous state features rather than calibrated speed dynamics.

Summary answers to key research questions:

Question	Answer
Is current aero mode detectable?	Yes — RF AUC = 1.000 from instantaneous features
Are driver anomalies detectable?	Yes — 5% rate, large effect sizes, GPS-mapped to technical corners
Can we predict 1s ahead on unseen circuits?	Yes — AUC-ROC 0.963 (LR) to 0.818 (CNN)
Does deep learning improve prediction?	No — all deep models perform below LR-instant
Why do deep models fail?	Circuit memorisation of absolute speed levels
Which deep model is most robust?	Transformer (AUC 0.911, Monaco 0.956)
What is the useful horizon?	≤ 1 second; all methods fail beyond 2.5 seconds
What are the most important features?	Throttle, Speed, nGear (instantaneous state)
What is the hardest test scenario?	Monaco street circuit + mode transition windows

7. Future Work

Addressing circuit memorisation (highest priority):

1. **Circuit-adaptive normalisation:** Replace absolute Speed with speed relative to the circuit's mean or maximum (computable from training data without leaking test information). This would eliminate the calibration anchor that causes GRU and RF-lag to fail at Monaco and is the most direct solution to the identified root cause.
2. **Domain-adversarial training:** Add a circuit-classifier adversarial head with gradient reversal, forcing the encoder to learn circuit-invariant representations. This approach has proven effective in domain adaptation problems analogous to the circuit-shift problem here.

Extending the evaluation:

1. **Broader circuit coverage:** The current LOCO benchmark covers 4 circuits. Extending to a full 24-race season would provide 24-fold evaluation with sufficient statistical power for Wilcoxon tests ($n \geq 10$ required for $p < 0.05$ two-tailed).
2. **Multi-driver study:** Current data covers two drivers; behavioural differences are statistically significant but small (Cohen's $d = 0.042$). A full-grid, multi-season study would test whether driver-specific models provide meaningful benefit.

Improving transition window performance:

1. **Transition-aware loss function:** Up-weight the ± 0.5 -second windows around mode changes in the focal loss objective. Table 8 shows these are the most safety-critical prediction events and the current loss function treats them identically to stable windows.

Deployment and outcome validation:

1. **Edge deployment latency:** All Phase 2 models are small (68k–213k parameters). Benchmarking inference latency on embedded hardware representative of F1-grade ECUs would confirm real-time feasibility.
 2. **Outcome-based label:** The current X-Mode label is a physics heuristic (fixed thresholds). A superior label would derive from lap-time simulations: "*would switching to X-Mode at this moment have improved lap time?*" This would align model objectives with engineering utility rather than rule recovery.
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